



FINANCIALTM
DATA EXCHANGE

Reference
Implementation
Docker Image
Retrieval for FDX API
v5.1



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Revision History

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Introduction

This document serves as a guide to set up and use the Reference Implementation application using a docker image. The instructions describe how to download and modify the default test data using Test Data Manager. Further instructions describe how to test endpoints with modified data via Postman.

The instructions comprise two stages:

1. Set up the environment and the application
2. Test the application

Application Setup

Prerequisites:

1. Docker Engine.
2. Docker-compose
3. The [ridocker.tar.gz](#) file.

Installation Steps:

1. Download the above [ridocker.tar.gz](#) file to the same server in which docker is installed.
2. **Untar** the contents using following command:

```
tar -xzf ridocker.tar.gz
```

Note: The tar command is supported in Windows 10 and above as well most Linux distributions. This results in the creation of two folders:

- a) proxy
 - b) ricopy
3. Determine if you need the Nginx reverse proxy for SSL or solely the backend application without SSL.
 - A. If you need Nginx:
 - i. Replace the certificates in the ssl folder with your private key and public key chain (certificate chain).
 - ii. Edit **default.conf** by replacing the <hostname> place holder with your domain name or IP address. This attribute occurs in three places within the conf file.
 - B. If you already have nginx or any other setup for routing HTTPS requests to backend port, please configure it to point to the RI backend by using the same settings from the **default.conf** file.

C. If HTTP and port 8080 is used then nginx is not required.

Ports used: 80, 443 by Nginx; 8080 by RI backend.

Note: If these ports are already used by another application, it may be necessary to change the port numbers. Files that would require port number edits are `docker-compose.yml` and `default.conf`.

Steps to bring up the server(s)

When you are ready to pull the images from our docker registry and start using the RI backend, perform the following steps:

1. Create the docker network that is required by docker-compose files using the command below.

```
docker network create fdxri_network;
```

You will receive the hash for the network you just created, though it is not required.

2. Change the directory to the RI backend folder as follows:

```
cd ricopy;
```

3. Login to the docker registry by using following command:

```
cat key/fdx_ri_copy.json | docker login -u _json_key --password-stdin https://gcr.io
```

You will be notified that you are successfully logged in. (Optionally you can configure credentials-store as shown by the warning or you can ignore the warning.)

4. Run the following command:

```
docker-compose up -d
```

This will pull the images and brings up the containers for you. **The RI backend server should be up and running.**

5. (Optional) If you need the nginx reverse proxy:
Go to the proxy folder:

```
cd proxy
```

Run the following commands:

```
docker-compose build
```

After a successful build:

```
docker-compose up -d
```

The nginx server is ready and configured to route http/https requests to RI backend.

Note: A docker-compose build will pull a nginx image and build a custom image as per the SSL settings. You will need to run this step the first time as well as every time you change the contents of proxy folder.

Access data manager using the following URLs:

1. If you are using Nginx or any custom setup to route SSL traffic to the backend RI:

```
https://<hostname>/adminui/fdx/datamanager
```

2. If you are not using any reverse proxy:

```
http://<hostname>:8080/adminui/fdx/datamanager
```

The default user name for test data manager is:

User name: admin

Password: admin

Testing

Once the Reference Implementation application is successfully installed, testing can be performed via the steps below. The application has two components:

1. FDX API
2. Test Data Manager (TDM)

FDX API Testing

Test an API Application set up using the following URLs:

1. If you are using Nginx or any custom setup to route SSL traffic to backend RI:

```
https://<hostname>fdxapi/accounts?resultType=lightweight
```

2. If you are not using any reverse proxy:

```
http://<hostname>:8080/fdxapi/accounts?resultType=lightweight
```

Note: Please replace <hostname> with your IP address or domain host name for above URL's

Test Data Manager

The application URL will begin with either https or http. This will be based on the Nginx reverse proxy settings: if set for SSL then use https, if without SSL then use http. **There will not be any auth/Bearer access token flow for the downloaded version. i.e., Auth Server and related APIs are not part of this docker image.**

This application facilitates the management of test data with a simple user interface. A user will land on a page which gives the option of downloading default data files or uploading modified data files for testing.

A. If you are using Nginx or any custom setup to route SSL traffic to a backend RI, use:

```
https://<hostname>/adminui/idx/datamanager
```

B. If you are not using any reverse proxy, use:

```
http://<hostname>:8080/adminui/idx/datamanager
```

The default user name for the test data manager is:

User name: admin

Password: admin

Note:

1. The filename should be the same as in the default set for the successful file upload.
2. The TDM also has a locking mechanism such that the user uploads a file and holds the lock for 4 hours or until the lock is released manually.
3. Users can revert the data to default data anytime by clicking the **Revert to default** button.
4. Additional users can be added using the /adminui/adduser API in the Postman collection (Under User management)

Test FDX API v4.6 & v5.1 via Postman collection

Perform the following to test via Postman:

1. Download the postman collection for v4.6 & V5.1 docker image [here](#).
2. Open the json file in a text editor.
3. Find the text <http://<hostname>:8080"> and replace it with your hostname / IP port according to your setup.
4. Import the collection to your Postman tool.

Note: For POST APIs Requests, wherever there is an idempotency key involved, the idempotency key should be unique for each new request. If an old idempotency key is sent with a new request, you will always receive the old response associated with the idempotency key.